

Generation Interconnection System Impact Study Report

For Queue Project Q21-1658

Donerville 138 kV
December 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1658. The Interconnection Customer proposes a 299.6 MW solar photovoltaic facility interconnecting at the Donerville 138 kV substation in the Pennsylvania area. The request was received on January 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,287,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rohrerstown to Gwf Tracy 138.0 kV line	Donerville to Rocky Mountain Hydroelectric Plant 138.0 kV	131.7	334	55.3
Gwf Tracy to Doneville 138.0 kV L	Donerville Substation to Rocky Mountain Hydroelectric Plant 138.0 kV	127.8	369	55.3
Irving Belt Line to Rd 138.0 kV	Donerville to Gwf Tracy 138.0 kV	122.6	236	37.6
Road to Rocky Mtn Hydroelectric Plant 138.0 kV	O'Fallon Renewable Energy Ctr to Culebra 3801 138.0 kV L	117.3	668	62.5
Rocky Mountain Hydroelecric Plant to Donerville 138.0 kV	Culebra 3801 to O'Fallon Renewable Energy Center 138.0 kV line	116.0	702	62.5
Engleside to Ipl Fit Monrovia I Llc 138.0 kV line	Culebra 3801 to O'Fallon Renewable Energy Center Substation 138.0 kV	115.4	431	41.6
Engleside to Milwood 138.0 kV	Culebra 3801 to O'Fallon Renewable Energy Ctr 138.0 kV	113.9	460	41.6
Rd to Millwood 138.0 kV	O'Fallon Renewable Energy Ctr to Culebra 3801 138.0 kV	112.1	497	41.6
Kingston to Ipl Fit Monrovia I Llc 138.0 kV Line	Conowingo 345-138 kV XFMR	111.6	261	11.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gwf Tracy to Rohrerstown 138.0 kV	Rebuild line with higher-capacity conductor	\$1,201,000	34% (\$403,000)
Gwf Tracy to Doneville 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,285,000	34% (\$1,773,000)
Irving Belt Line to Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$1,756,000	27% (\$477,000)
Road to Rocky Mtn Hydroelectric Plant 138.0 kV	Rebuild line with higher-capacity conductor	\$3,088,000	24% (\$735,000)
Rocky Mtn Hydroelectric Plant to Donerville 138.0 kV	Rebuild line with higher-capacity conductor	\$1,945,000	24% (\$463,000)
Engleside to Ipl Fit Monrovia I Llc 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,292,000	22% (\$515,000)
Engleside to Milwood 138.0 kV	Rebuild line with higher-capacity conductor	\$1,321,000	22% (\$297,000)
Millwood to Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$3,065,000	22% (\$688,000)
Kingston to Ipl Fit Monrovia I Llc 138.0 kV Line	Rebuild line with higher-capacity conductor	\$8,914,000	22% (\$1,936,000)
	Total allocated to project		\$7,287,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.